Paper 5.75 - Oloid Path Carrier as Next-State Precondition

CQE-paper-05.75
CQECMPLX Formal Paper Series
Review-facing PDF built 2026-06-12

Construction Basis

This PDF is the clean paper artifact. Source extracts, kernels, receipts, tool notes, workbook material, and package bindings are used as construction evidence; they are not treated as separate review-facing papers.

Evidence Inputs

- top-level review source: Papers\Source\CQE-paper-05.75.md

Paper 5.75 - Oloid Path Carrier as Next-State Precondition

Purpose

Paper 5.75 explains how the rolling path carrier becomes a precondition for Paper 6 and later accumulated-center papers.

Exported State

Paper 5 exports:

```
rolling trace
legal adjacent-step relation
head/tail dyads
payload attachment rule
invalid-bit rejection
discontinuous-jump rejection
scope boundary: no prediction claim
```

Use in Paper 6

Paper 6 may register Paper 5 outputs as causal proof edges. The edge must preserve:

```
source paper
target paper
edge type
receipt
status
```

Paper 5 proves path continuity. Paper 6 proves that the dependency graph records that continuity without hiding obligations.

Use in Paper 7

Paper 7 may sample or visualize a carried trace only if it preserves the original discrete receipt. Interpolation or presentation cannot erase the rolling trace that Paper 5 proves.

Use in Later Papers

Later papers may use Paper 5 as the carrier for:

- accumulated center state,
- Hamiltonian window inputs,
- mass-residue carriers,
- lattice-boundary payload transport,
- product and adapter workflows.

Every use must state whether it imports the finite carrier theorem or proves a new prediction/physics claim.

Precondition Rule

Before a later paper uses Paper 5, it should be able to answer:

```
What is the input stream?
What is the current rolling state?
```

```
Which adjacent step proves continuity?
What payload is carried?
Does the payload alter the path rule?
Which receipt proves the carrier fact?
```

Conclusion

Paper 5.75 turns the rolling path theorem into portable state. It exports a legal carrier that can move repaired constraints forward while keeping prediction and physical interpretation tied to their own later receipts.